Influence of Internal Flow on Film Cooling Effectiveness

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Film cooling experiments were conducted to investigate the effects of internal flow conditions and plenum geometry on the film cooling effectiveness. The film cooling measurements show a strong influence of the coolant inlet conditions on film cooling performance. The present experiments were carried out on a flat plate with a row of cylindrical holes oriented at 30 deg with respect to a constant-velocity external flow, systematically varying the plenum geometry and blowing rates ($0.5 \leqslant M \leqslant 1.25$). Adiabatic film cooling measurements using the multiple narrow-banded thermochromic liquid crystal technique (TLC) were carried out, simulating a flow parallel to the mainstream flow with and without crossflow at the coolant hole entry compared with a standard plenum configuration. An impingement in front of the cooling hole entry with and without crossflow was also investigated. For all parallel flow configurations, ribs were installed at the top and bottom coolant channel wall. As the hole length-to-diameter ratio has an influence on the film cooling effectiveness, the wall thickness has also been varied. In order to optimize the benefit of the geometry effects with ribs, a vortex generator was designed and tested. Results from these experiments show in a region $5 \leqslant X/D \leqslant 80$ downstream of the coolant injection location differences in adiabatic film cooling effectiveness between +5 percent and +65 percent compared with a standard plenum configuration.
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Introduction

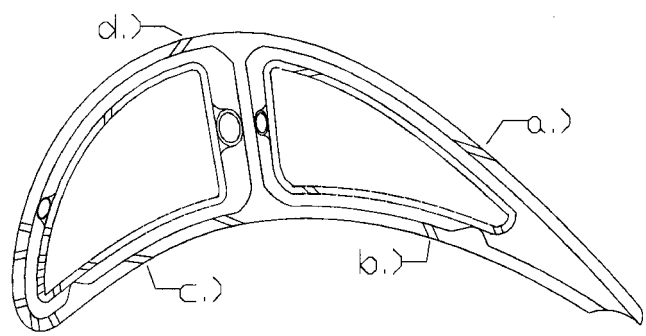
Rising turbine inlet temperatures have led to extensive use of film cooling technology and research work within this area. The review by Goldstein [1] delineates much of the physics of film cooling, especially the jet in crossflow behavior. In film cooling literature a large number of parameters have been investigated with numerical and experimental methods, for example, Ardey and Fottner [2], Chernobrovkin and Lakshminarayana [3], Leylek and Zerkle [4], Papell [5], Sinha et al. [6], and Walters and Leylek [7] but most of them used a classical plenum for hole supply. Modern cooled turbine vane (Fig. 1) cooling configurations combine convective impingement and film cooling. Therefore, a classical plenum configuration is hardly used, but hole configurations supply parallel flow at the cooling hole entrance. The inlet flow conditions to the film cooling holes are different for nearly every hole position.

Wilfert [8] and Vogel [9,10] investigated vortex generation and mixing behavior on a scaled airfoil at real turbomachinery flow conditions. Their experimental and numerical results showed a significant influence of inner contouring on mixing behavior and film cooling performance. Thole et al. [11], Wittig et al. [12], and Gritsch et al. [13] varied hole supply Mach number, flow angle, hole geometry, and inclination angle, and showed a significant influence on the hole exit and downstream flow field and on the film cooling performance. They used a two-dimensional coolant channel with $H/D=2$. Berhe and Patankar [14] found an influence of the internal flow for $H/D \leqslant 2$. Burd and Simon [15] determined differences up to 20 percent in lateral averaged adiabatic effectiveness between a plenum and a parallel crossflow configuration ($H/D=2$; $VR=1$).

The present study is based on a development that intensifies the positive effects of parallel cross flow shown in Fig. 2 with ribs at the top and bottom wall of the coolant supply channel. In order to get valuation criteria for actual hole supply configuration designs, an investigation on four different internal flow configurations, and a classical plenum as a reference were carried out, systematically measuring adiabatic film cooling effectiveness. Low and high turbulent coolant flows were simulated with and without crossflow. Additionally the length of the film cooling hole was varied.

Experimental Apparatus

The adiabatic film cooling experiments were conducted in a subsonic open-circuit wind tunnel with a straight channel test sec-



a.) parallel & cross flow
b.) parallel & no cross flow
c.) impingement & cross flow
d.) impingement & no cross flow

Fig. 1 Cross section of a combined cooled turbine blade

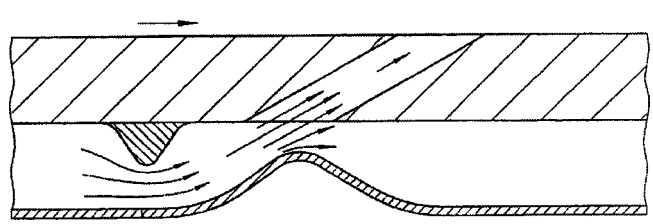


Fig. 2 Internal geometry for improved vane cooling

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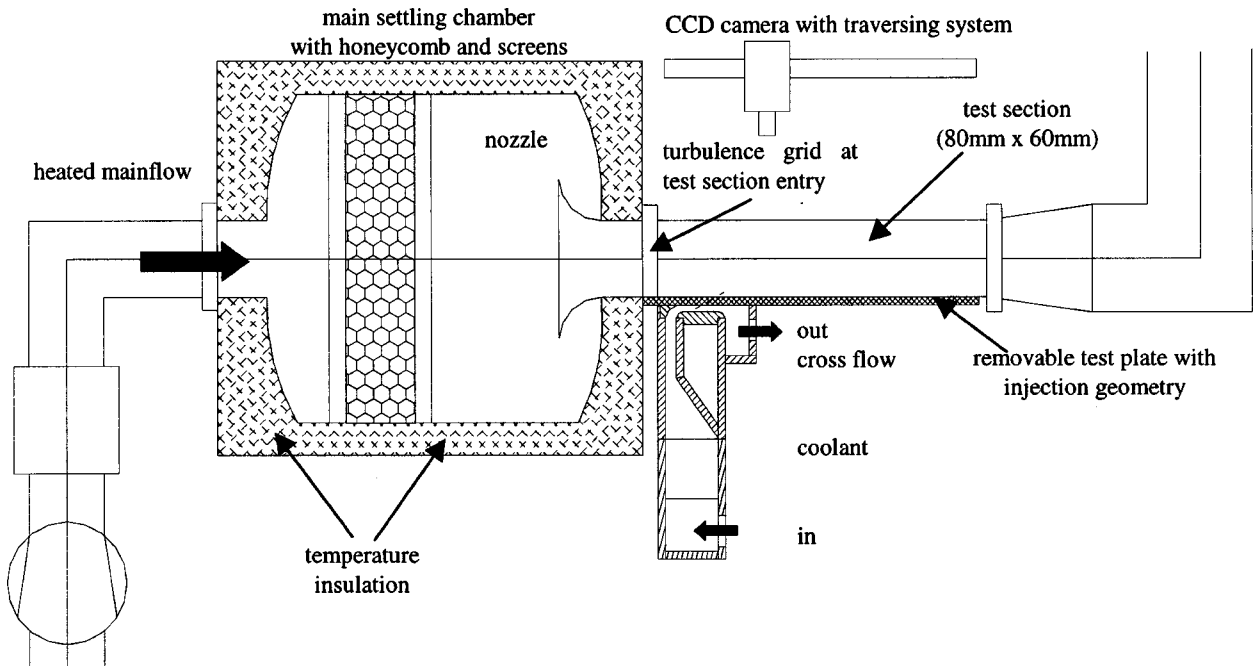

Fig. 3 Straight channel test facility

tion shown in Fig. 3. Ambient air was compressed by an electrical blower. Free-stream air was heated in a range of 60°C to 70°C by an electrical heater before it entered the settling chamber. The hot air path and the settling chamber were insulated in order to minimize temperature differences of the incoming flow. An assembly of honeycomb and a series of screens homogenized the flow inside the settling chamber. The flow accelerated through the nozzle with a 10:1 contraction ratio into the test section.

A turbulence grid was placed at the test section entrance to increase the free-stream turbulence intensity. This turbulence grid consisted of 3×3 mm rectangular holes that were spaced 4 mm in width and height from one hole center to the next hole center. The streamwise decay of the free-stream turbulence intensity was determined in previous measurements on this test facility ($Tu_\infty = 5.0$ percent at $X/D=18$, $Tu_\infty = 3.5$ percent at $X/D=0$, and $Tu_\infty = 2.0$ percent at $X/D=40$). The approaching boundary layer was turbulent with a boundary layer thickness of about 4 mm at the injection location. The displacement thickness at this location was determined to be 0.5 mm. The free-stream velocity level was about $U_\infty = 115$ m/s.

A second pressurized air supply system of the lab delivered the cooling mass flow at an ambient temperature level of about 20°C. The incoming cooling mass flow was accurately determined by means of a turbine wheel meter (1 percent measurement uncertainty) and piped into the cooling air settling chamber, where the air passed through a set of flow-straightening devices to minimize flow disturbances. Thereafter, the coolant flow was accelerated into a vertical channel of $H/D=3$ and turned into horizontal direction by a bend with another contraction of the channel to $H/D=2$. Ribs are placed on the top and bottom side of the coolant passage (Fig. 5(*a*)) in order to optimize the inflow into the cooling holes. The cooling air was injected into the heated free-stream flow through a single row of cylindrical holes at selected blowing rates in the range of $0.5 \leqslant M \leqslant 1.25$. To simulate crossflow conditions at the hole entry, a part of the mass flow was sucked off at the end of the coolant channel and measured by means of a second turbine wheel. The injected cooling mass flow was determined by the difference between the incoming and outgoing mass flow with an accuracy of better than 4 percent.

The typical density ratio for all experiments conducted was DR=1.15. For the impingement configurations, the bend was replaced by a row of 10 holes ($D_i = 1.25D$; $P_i/D_i = 2$) to impinge the cooling gas $20D$ upstream on the top wall of the coolant channel (Fig. 4, right side).

The dimensions of the hot gas flow channel in the straight test section were 80 mm in width and 60 mm in height with top and side walls constructed of polycarbonate material. The polycarbonate walls allowed optical access to the test surface. A color CCD camera was arranged to observe the test surface through the top wall of the test section. Secondary air was injected after 90 mm of the total test section length of 600 mm. The bottom wall of the straight channel, the actual test surface, held the different injection configurations investigated. The use of low conductivity material (Necurom, $k = 0.1$ W/m/K) and additional thermal insulation applied below this test surface kept the heat losses to low values. Temperature measurement errors due to heat losses were determined to be less than 0.05°C by a two-dimensional heat conduction analysis.

Figure 5 shows the top and side view of the removable test plate and illustrates the injection configurations. All configurations consisted of a single row of five cylindrical holes. The hole spacing in lateral direction corresponded to a pitch-to-diameter ratio of $P/D=4$. The hole channels were inclined at $\alpha=30$ deg with respect to the free-stream flow direction. The hole length-to-diameter ratio of the injection channels was varied from $L/D=4$ (Fig. 5(a)) to $L/D=8$ (Fig. 5(b)), by increasing the test plate thickness from 8 to 16 mm.

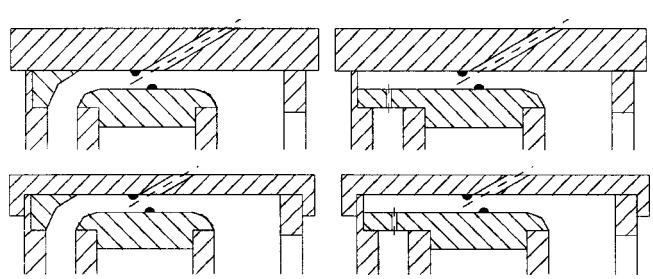

Fig. 4 Injection configurations

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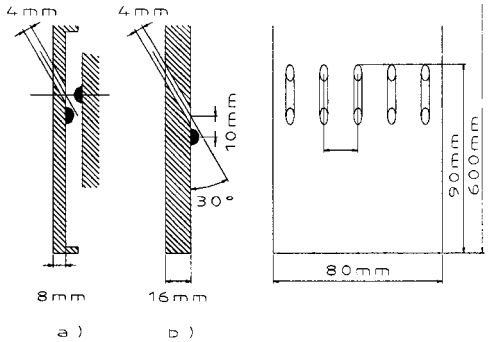

Fig. 5 Injection configurations ($P/D=4$): (*a*) $L/D=4$; (*b*) $L/D=8$

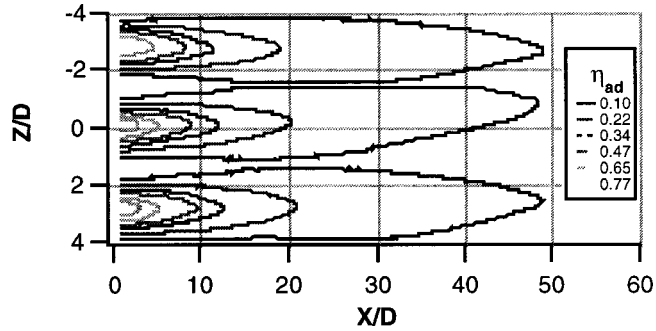

Fig. 6 Typical TLC contours of film cooling effectiveness values at a low blowing rate

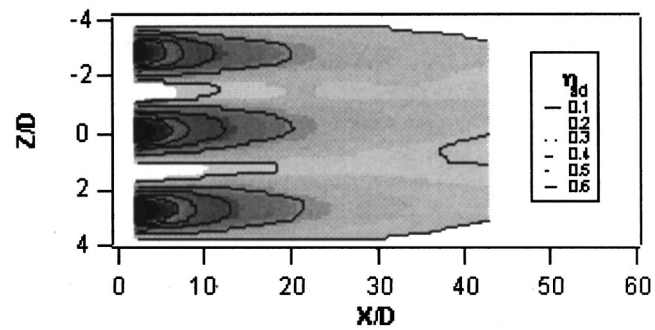

Fig. 7 Typical interpolated film cooling effectiveness data at a low blowing rate

Data Analysis

Adiabatic film cooling data were obtained using the well-established thermochromic liquid crystal technique (TLC) to measure the adiabatic wall temperature distribution. Isothermal temperature patterns due to film cooling on the adiabatic test plate were indicated by a mixture of four narrow-banded thermochromic liquid crystals and were recorded by a color CCD camera in conjunction with a color image processing system. All additional experimental data, i.e., temperature, pressure, and mass flow information of the main and secondary flow, was stored by a separate data acquisition system for later data reduction purposes.

To achieve an accurate measurement of the wall temperature using the TLC's, a special apparatus was built to conduct temperature calibration of the TLC indication temperatures inside the film cooling test section. The calibration was conducted by interchanging the test plate with a copper calibration plate. A linear temperature distribution was achieved by heating one end of this plate and cooling the other end. Determination of the TLC temperature indications on the copper surface was obtained with the same color image processing system used for the film cooling measurements. Thermocouples, embedded in the copper surface, were used to measure the temperature distribution over the plate length. The experimental uncertainty of the local film cooling effectiveness values was determined by an uncertainty analysis to be less than 0.01 [16]. To investigate the influence of the backside flow on the test surface, two-dimensional heat flux calculations were performed. Only for the results up to $X/D=7$ for the thin test surface ($L/D=4$) was an influence of the increased convective heat flux due to the coolant crossflow determined.

During the experiments, the TLC data were captured using a color CCD camera in conjunction with a color image processing system. A complete data set was compiled using up to six adjacent smaller images. The camera was mounted above the experimental apparatus on a traverse system controlled by a stepping motor, allowing all pictures to be taken with the same reference points. Using multiple images to build up the final picture allowed higher resolution and better isothermal definition of the adiabatic wall temperature distributions. The temperature indications of the TLC's were assumed to be the adiabatic wall temperatures. The free-stream temperature was measured by a total temperature probe at $X/D=-10$ upstream of the injection location. The coolant temperature was measured by a thermocouple in the coolant settling chamber. With the additional information of the coolant and free-stream temperatures, these isothermal contour lines were recalculated to adiabatic film cooling effectiveness contours as shown in Fig. 6 by the following equation:

$$\eta_{ad}=\frac{T_\infty-T_{\mathrm{TLC}}}{T_\infty-T_c} \tag{1}$$

For the calculation of the lateral averaged film cooling effectiveness values, a special interpolation technique was used. The field of interpolation was determined over a range of two pitches (three holes) in the lateral direction. In the streamwise direction it was depending on the input data set, respectively (Fig. 7). For details see Lutum and Johnson [16].

Finally, an integration in the lateral direction was performed to determine the lateral averaged film cooling effectiveness data at each streamwise position. The estimated integration error based on the experimental and interpolation uncertainty was about 1 to 3 percent of the streamwise local lateral averaged film cooling effectiveness. The uncertainty depends on the effectiveness level, with the higher uncertainty corresponding to lower effectiveness values.

Present Results

The test program comprises five configurations (Table 1) with different inlet flow conditions. Configuration 1 is a classical plenum. where the coolant is sucked in from a chamber, whereas the other configurations have a supply channel parallel to the mainstream direction. For configurations 2 and 3 a bend is used to direct the coolant in horizontal direction, whereas for configurations 4 and 5 the coolant impinges on the upper wall of the coolant supply channel. To simulate crossflow conditions (configurations 2 and 4), a part of the cooling gas is sucked off downstream of the supply channel. The crossflow Mach numbers inside of the supply channel ($\mathrm{Ma}_{c,cr}$) upstream and downstream of the cooling hole entrance are given in Table 1. The various blowing ratios were investigated for $L/D=4$ and $L/D=8$. Additional investigations were performed without guiding ribs.

In Fig. 8 the results for $L/D=8$ and $L/D=4$ of configuration 1 (plenum) are compared with the results of Lutum and Johnson [16]. The film cooling effectiveness is in the expected range considering that Lutum and Johnson used a film cooling configuration with an inclination angle $\alpha=35$ deg and a blowing rate of $M=1.15$ (configuration 1, $\alpha=30$ deg, $M=1.0$). Detailed validation of the test facility can be found in Lutum and Johnson [16].

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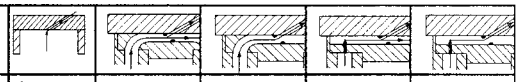
Table 1 Measurement program

Sketch					
Nr.:	Conf. 1	Conf. 2	Conf. 3	Conf. 4	Conf. 5
$Ma_{c,cr}$ up/ downstream	0.00 0.00	0.12 0.08	0.02 0.00	0.12 0.08	0.02 0.00
M =	0.5 – 1.25				
Ribs	-	With & without ribs			
L/D =	8 & 4				

Measurements of adiabatic film cooling effectiveness for selected hole supply configurations ($L/D=8$), are presented in Figs. 9–14. All results are shown with the dimensionless streamwise coordinate X/D. Configuration 4 with a simulated impingement flow and a crossflow is used to show the differences to the plenum configuration. The film cooling effectiveness distributions for these two configurations are shown in Figs. 9 and 10. The adiabatic film cooling distribution for configuration 1, given in Fig. 9 ($L/D=8$), shows a typical pattern of effectiveness with high values just behind the injection location and a lateral spreading of the cooling film further downstream. The distribution of the impingement with crossflow configuration (configuration 4) indicates a higher maximum value of 0.5 up to $X/D=9$ and larger areas in lateral and streamwise directions with effectiveness values over 10 percent. However this kind of graphic presentation is not useful when attempting to create quantitative performance comparisons between different film cooling setups. Therefore, the lateral averaged film cooling effectiveness is used to appraise all configurations. These values and variations from a reference case give an easy way to compare different configurations.

The plenum configuration (Fig. 11) shows rapidly decreasing film cooling effectiveness downstream of the injection location for $M=0.5$. The higher blowing ratios start with lower effectiveness, but level off at higher values further downstream, especially at $M=1.0$ and $M=0.75$.

The impingement with crossflow configuration (Fig. 12) shows the same performance for $M=0.5$, but different progressions for higher blowing ratios. The effectiveness near the hole is higher for $M=1.0$ and $M=1.25$, and for $M=1.0$ the decrease in effectiveness in streamwise direction is less rapid than for the plenum configuration. All configurations level off at higher effectiveness at $X/D=80$.

Comparisons of the tested configurations are represented by the improvement of lateral averaged film cooling effectiveness, setting the plenum configuration at 100 percent as a reference.

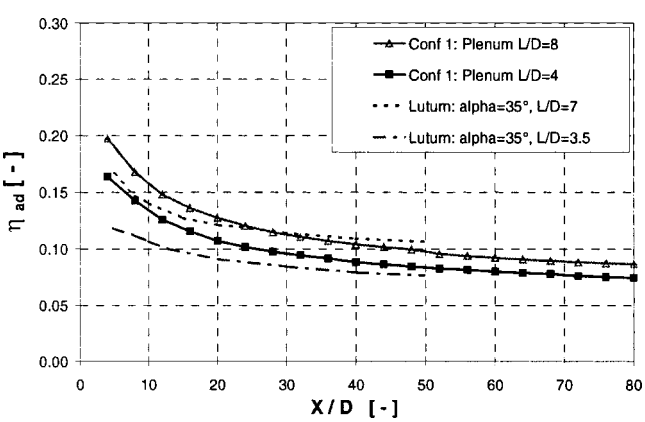


Fig. 8 Lateral averaged film cooling effectiveness for configuration 1 ($M=1.0$) and Lutum ($M=1.15$)

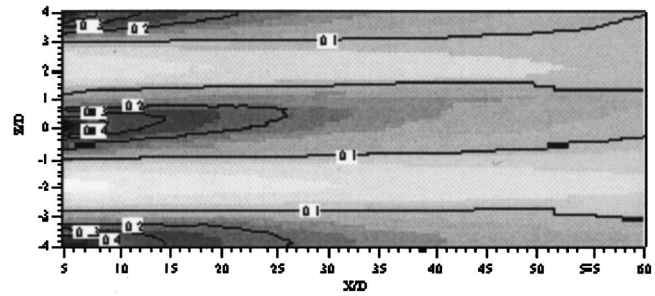


Fig. 9 Adiabatic film cooling effectiveness distribution for $M=1$, configuration 1 ($L/D=8$)

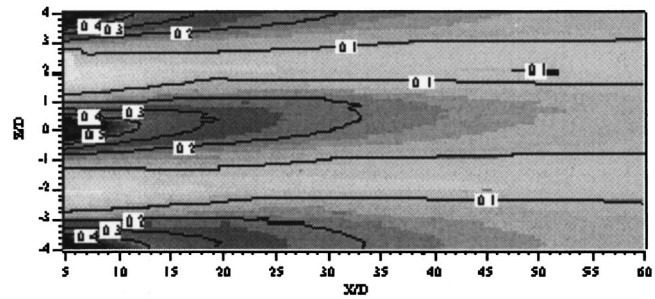


Fig. 10 Adiabatic film cooling effectiveness distribution for $M=1$, configuration 4 ($L/D=8$, with ribs)

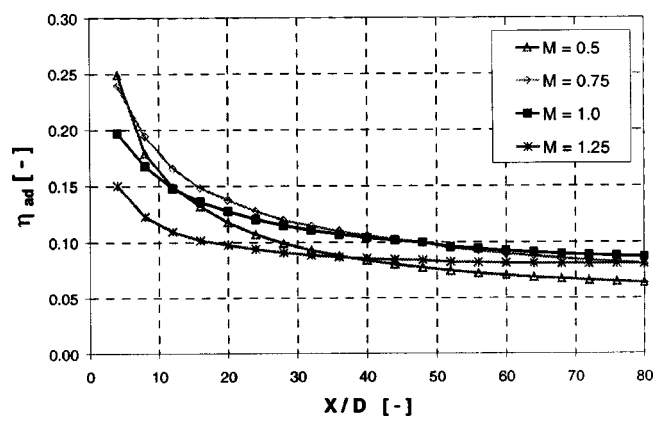


Fig. 11 Influence of blowing rate on lateral averaged adiabatic film cooling effectiveness for configuration 1 ($L/D=8$)

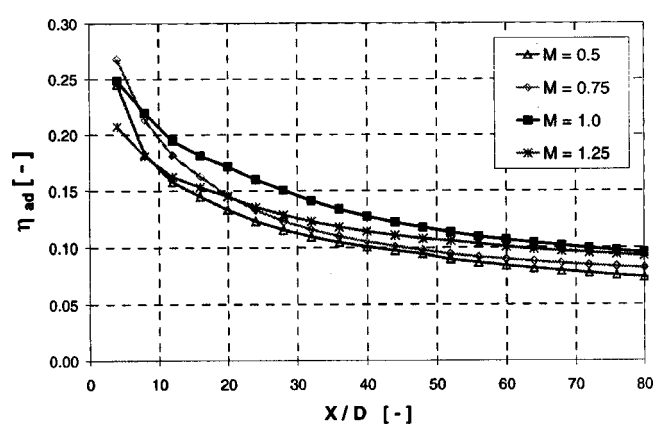


Fig. 12 Influence of blowing rate on lateral averaged adiabatic film cooling effectiveness for configuration 4 ($L/D=8$, with ribs)

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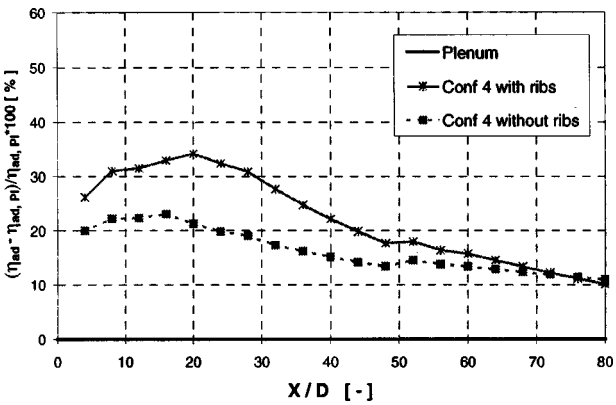

Fig. 13 Influence of guiding ribs on improvement of lateral averaged adiabatic film cooling effectiveness ($M=1$, $L/D=8$)

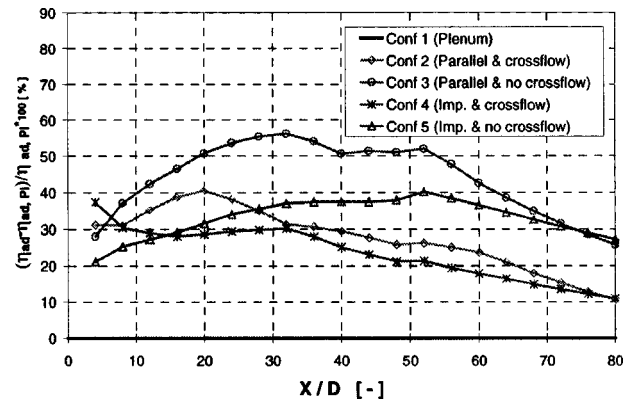

Fig. 15 Improvement of lateral averaged adiabatic film cooling effectiveness ($M=1$, $L/D=4$, with ribs)

$$\text{Improvement}=\frac{\eta_{ad}-\eta_{ad\ \mathrm{Pl}}}{\eta_{ad\ \mathrm{Pl}}}*100\ \ [\text{percent}] \tag{2}$$

To be able to separate the influence of internal flow parallel to the mainstream and the ribs for guiding the flow into the film cooling hole, tests were conducted without ribs. Figure 13 shows the improvement of film cooling effectiveness for configuration 4 with and without ribs at the blowing rate $M=1.0$. For the near-hole field, an increase of about 10 percent was measured for the configuration with ribs.

The improvement in lateral averaged film cooling effectiveness for configurations 2–5, given in Fig. 14, shows less than 10 percent differences between the plenum configuration and the configurations with no cross flow. Nevertheless, for the crossflow configurations, an improvement up to 34 percent is determined. Furthermore, for the impingement with crossflow configuration (configuration 4), an improvement of about 22 percent up to $X/D=80$ in average can be observed. The increasing improvement up to $X/D=20$ may be caused by an additional pair of vortices [17]. The pair of vortices is conserved further downstream than the kidney-shaped vortex and is transporting the hot gas through the jet to the surface and not sucking the hot gas between the holes to the surface and underneath the jet.

Effects of the internal flow decrease for longer holes. Therefore, measurements were performed for shorter holes at $L/D=4$. In Fig. 15 the improvements for configurations 2–5 are shown again referenced against the plenum configuration. As expected, the influence is increasing for shorter holes. Configuration 4 with impingement and crossflow has about the same effectiveness as for $L/D=8$. All other configurations show an improvement of more than 30 percent. Especially configurations 3 and 5 without crossflow now have the highest values. For configuration 3 an improvement up to 56 percent maximum and 43 percent in average is determined. For the long holes, the influence of the ribs was also investigated. Figure 16 shows the influence for configuration 3. For this case without crossflow, the improvement with the ribs is about 30 percent up to $X/D=40$ compared to the case without ribs.

These results lead to the conclusion that the exit flow distribution and its vortices have a strong influence on the mixing behavior and therefore on the film cooling effectiveness. The above-mentioned counterrotating vortices measured by Wilfert [17] are very stable and have a positive influence on the mixing. To intensify this pair of vortices, a prevortex generator was developed and tested (Fig. 17). The vortex generator uses the mechanism of secondary flow production. If the coolant flow is redirected from horizontal direction into the cooling holes, a pressure gradient is induced. The radial equilibrium for the flow in a bend is defined as follows:

$$\frac{\partial p}{\partial r}=\rho_\infty\frac{U^2}{R_\infty} \tag{3}$$

The interaction of pressure gradient and the side wall boundary layer of the vortex generator results in a counter-rotating pair of vortices inside the hole.

The film cooling effectiveness distribution for configuration 3 with ribs and with vortex generator is shown in Figs. 18 and 19. The values on the centerline are higher for the case with the vor-

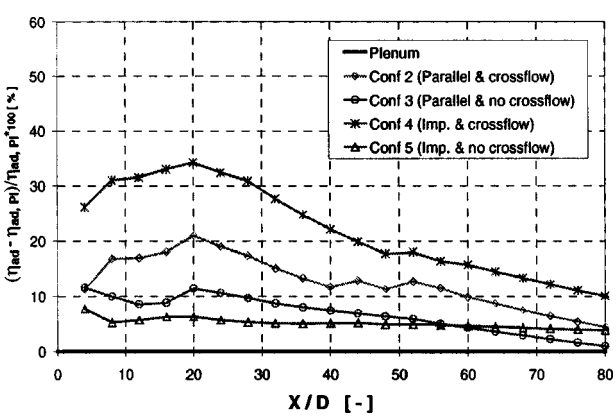

Fig. 14 Improvement of lateral averaged adiabatic film cooling effectiveness ($M=1$, $L/D=8$, with ribs) configurations ($M=1$, $L/D=8$)

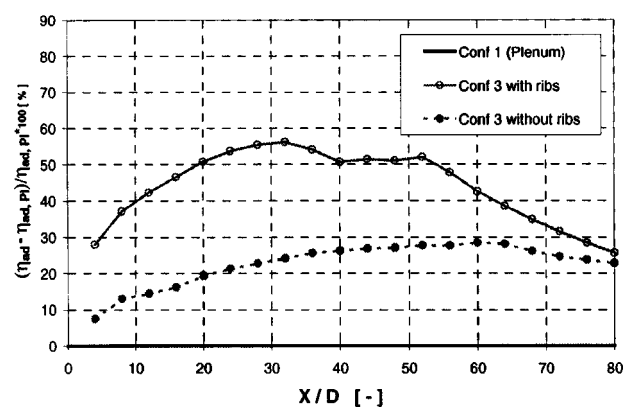

Fig. 16 Influence of guiding ribs on improvement of lateral averaged adiabatic film cooling effectiveness ($M=1$, $L/D=4$)

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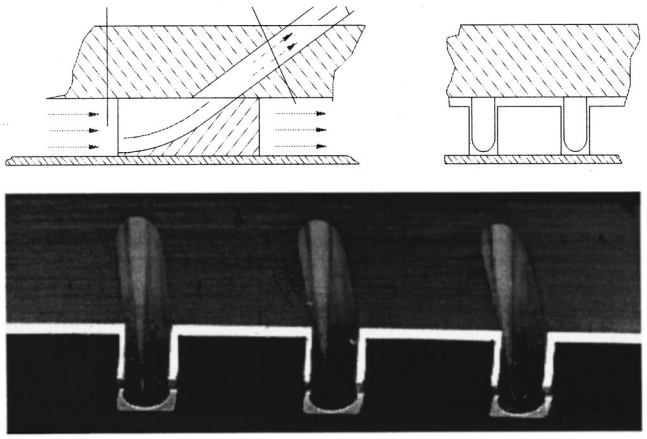

Fig. 17 Sketch and photo of vortex generator setup

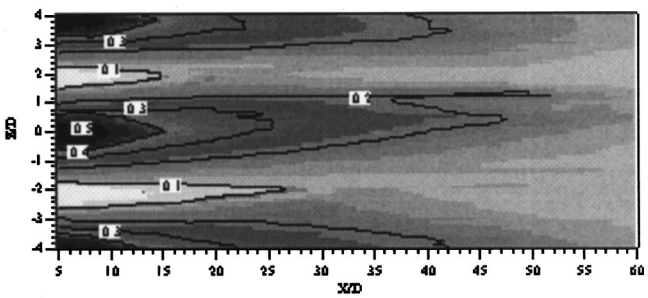

Fig. 18 Adiabatic film cooling effectiveness distribution for $M=1$, $L/D=4$, configuration 3 with vortex generator

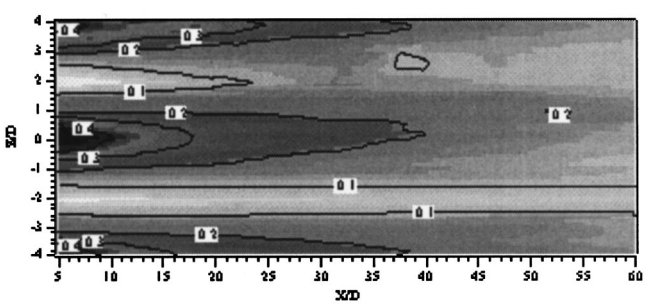

Fig. 19 Adiabatic film cooling effectiveness distribution for $M=1$, $L/D=4$, configuration 3 with ribs

tex generator and the low effectiveness regions between two holes are smaller. The improvement of lateral averaged adiabatic film cooling effectiveness to the configuration with vortex generator, as well as the best configuration with parallel flow and ribs, shown in Fig. 20, are again referenced to the plenum configuration. The benefit of the vortex generator is quite obvious, improvements above 80 percent maximum value and above 65 percent in average up to $X/D=80$ are determined compared to the standard plenum configuration. Furthermore, an improvement up to 15 percent in average between configurations with the vortex generator and the best configuration with ribs could be observed.

Conclusions

The present measurements demonstrate the influence of the inner contouring of the coolant channel and the hole supply on lateral averaged film cooling effectiveness. Compared to plenum configurations, deviations in adiabatic film cooling effectiveness

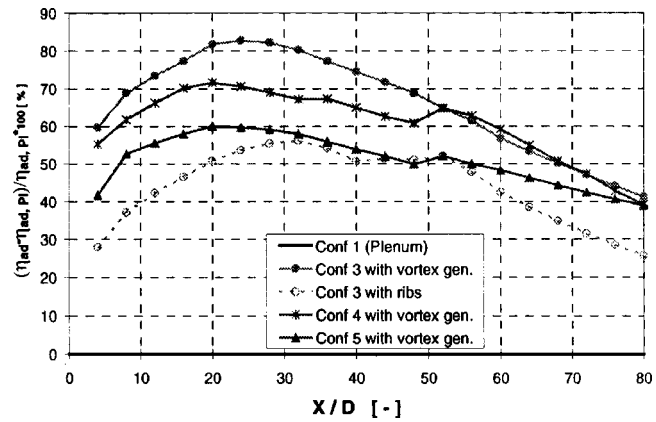

Fig. 20 Improvement of lateral averaged film cooling effectiveness for configurations 3, 4, and 5 with vortex generator and configuration 3 with ribs ($M=1$, $L/D=4$)

of up to 30 percent can be observed. The lateral averaged film cooling effectiveness increases up to 56 percent maximum and 43 percent in average when using guiding ribs to redirect the coolant into the holes. Furthermore, some interesting observations are made:

- Plenum geometry and hole supply have a stronger influence for low L/D ratios than for high L/D ratios.
- For high L/D ratios, the coolant channel velocity (crossflow) has a strong influence whereas the guiding ribs have just a small effect.
- Small L/D ratios show a different trend: Small influence of coolant crossflow and a strong influence of the redirection by means of the guiding ribs.

In order to optimize the benefit due to the geometry effects on the lateral averaged film cooling effectiveness, a vortex generator was designed and tested.

- The vortex generator leads to better lateral spreading of the cooling gas on the cooled surface.
- The vortex generator shows the best results for lateral averaged film cooling effectiveness of all tested configurations with an improvement of up to 83 percent maximum and about 65 percent in average compared to the classical plenum configuration for an internal parallel flow without crossflow.

To get usable predictions for modern film cooled turbine vanes, the plenum geometry and the hole supply have to be taken into account for future film cooling investigations. Very often cylindrical holes are used at high blowing rates instead of shaped holes because shaped holes have higher manufacturing costs. The effects presented for $M=1$ are increasing with increasing blowing rate. Especially for higher blowing rates, an important improvement in film cooling effectiveness can be achieved with optimized internal flow conditions.

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Nomenclature

D = coolant hole diameter, m
DR = density ratio $=\rho_c/\rho_\infty$
H = coolant channel height, m
H/D = dimensionless height of coolant channel
VR = velocity ratio $=U_c/U_\infty$

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I = momentum flux ratio$=\rho_c U_c^2/\rho_\infty U_\infty^2$
L = length of coolant passage, m
M = blowing rate$=\rho_c U_c/\rho_\infty U_\infty$
P = hole pitch, distance between holes in lateral direction, m
P/D = dimensionless lateral spacing, pitch to diameter ratio
R = radius, m
T = temperature, °C
Tu = turbulence intensity, percent
U = mean velocity, m/s
X = streamwise coordinate, origin at film hole center, m
X/D = dimensionless streamwise coordinate
Z = lateral coordinate, m
Z/D = dimensionless coordinate in lateral direction
k = thermal conductivity, W/m/K
α = injection angle of the coolant fluid, deg
$\eta_{ad,la}$ = lateral averaged adiabatic film cooling effectiveness
ρ = fluid density, kg/m^3

Subscripts

c = coolant fluid
cr = crossflow
i = impingement
∞ = free-stream fluid
Pl = plenum

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